The Lifetime of Cosmic-Ray Muons Stopped in Lead Glass

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Cosmic-ray muons stopped in lead glass were timed from their arrival to the light of a decay electron in four independently configured acquisitions. Each spectrum contains a narrow secondary peak tentatively attributed to photomultiplier afterpulsing. Fits exclude this peak and retain the early decay where no extra peak is visible. Two acquisitions pass both a single-exponential shape test and a tail-window stability test. Their weighted mean gives $\tau = (2.19 \pm 0.04\ (\text{stat + scatter}) \pm 0.06\ (\text{sys}))$ µs using the 1.28 µs calibration employed in the experiment; the scatter term allows for their disagreement. This agrees with the free-muon lifetime of 2.197 µs, but lies 0.09–0.15 µs above the 2.04–2.10 µs calculated for capture of negative muons in the assumed glass. The calibration was recorded on a different analyser input from the higher-weight acquisition, so a small scale difference between inputs cannot be excluded. That possibility and the capture model's assumptions limit the interpretation as a free-muon lifetime.

1. INTRODUCTION

The mean lifetime of the muon, 2.197 µs for positive muons [1], can be measured on a laboratory bench with cosmic rays. A muon that stops in a block of lead glass produces a flash of light, and the electron from its decay produces a second flash a few microseconds later. Converting the delay between the two flashes into a pulse height and histogramming many such delays gives a distribution that falls exponentially with the lifetime, provided every delay is detected with the same efficiency.

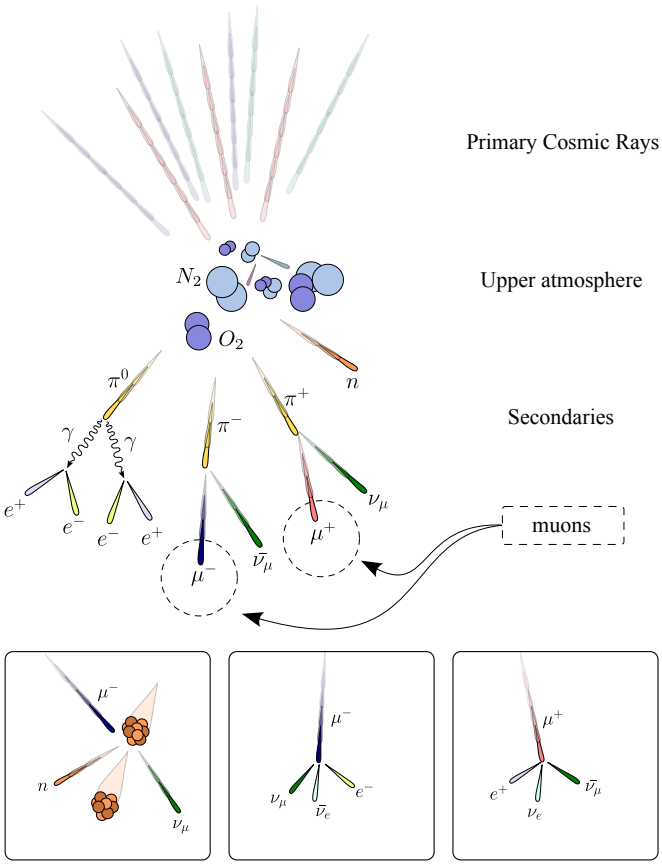
In practice the photomultipliers produce delayed pulses of their own, and the electronics record unrelated coincidences. This report asks whether a stable decay constant can be extracted once these effects are identified and excluded. Four acquisitions made with different detector settings were analyzed separately, and their agreement serves as a test of the result.

2. PHYSICAL BASIS

2.1 Production and decay of cosmic-ray muons

Primary cosmic rays, mostly protons and heavier nuclei, collide with nitrogen and oxygen nuclei in the upper atmosphere and produce showers of secondary particles (Fig. 1, upper panel). Charged pions in these showers decay through $\pi^+ \to \mu^+ + \nu_\mu$ and $\pi^- \to \mu^- + \overline{\nu}_\mu$, and these muons make up most of the charged particles that reach the ground. Neutral pions instead feed the electromagnetic part of the shower. A muon has about 207 times the mass of an electron, so it loses far less energy to radiation and can pass through the whole atmosphere [2].

Some of these muons stop inside the lead glass. A stopped positive muon can only decay, producing a positron and two neutrinos (Fig. 1 c). A stopped negative muon can also decay, to an electron and two neutrinos (Fig. 1 b), but it may instead be captured by a nucleus in the glass (Fig. 1 a). In either decay the electron or positron shares the available energy with the two neutrinos, so it emerges with a continuous energy spectrum



(a) muon capture (b) muon decay (c) anti-muon decay

FIG. 1. Production and fate of cosmic-ray muons, illustrated by the author. Upper panel: a primary cosmic ray strikes a nitrogen or oxygen nucleus; charged pions produce the circled muons and neutrinos, and neutral pions produce photons and electron–positron pairs. The altitude is schematic, since production occurs over a range of heights. Lower panels: processes available once a muon has stopped. (a) A negative muon can be captured by a nucleus before it decays. (b) A negative muon decays to an electron and two neutrinos. (c) A positive muon decays to a positron and two neutrinos. Only the charged products in (b) and (c) produce the delayed light signal; the nuclear products in (a) are schematic.

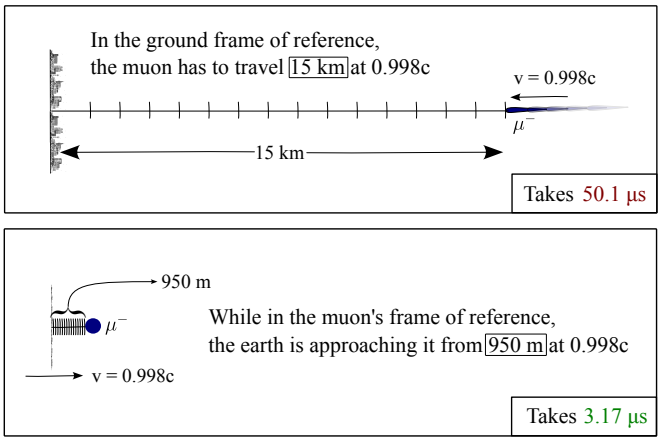


FIG. 2. The same muon journey in two inertial frames, illustrated by the author. The ground frame assigns a 15 km path and a dilated lifetime. In the muon's frame, Earth approaches across a contracted distance of about 950 m and the muon keeps its rest lifetime. Both frames give the same elapsed proper time and survival probability. The speed is fixed at 0.998c.

extending to about 53 MeV [2]. This charged particle produces the delayed light pulse that the experiment detects, while the neutrinos escape unseen.

2.2 Exponential decay and survival through the atmosphere

If a muon has survived to a given moment, its probability of decaying in the next short interval $\mathrm{d}t$ is $\mathrm{d}t/\tau$, independent of its age, where τ is the mean lifetime [3]. Among N muons the expected number decaying in $\mathrm{d}t$ is then $N\,\mathrm{d}t/\tau$, so

$$\frac{\mathrm{d}N}{\mathrm{d}t} = -\frac{N}{\tau}, \quad N(t) = N_0 \exp\!\left(-\frac{t}{\tau}\right), \tag{1}$$

where N_0 muons are present at $t = 0$.

On its own, Eq. (1) predicts that almost no muons produced high in the atmosphere reach the ground. Even at nearly the speed of light, a muon travels only about 660 m in one lifetime $\tau_\mu \approx 2.197$ µs, while the journey spans several kilometres. Time dilation accounts for the difference. For speed $v = \beta c$, a path L takes L/v on a ground clock but only $L/(\gamma v)$ on the muon's clock, where $\gamma = (1-\beta^2)^{-1/2}$ is the Lorentz factor [4]. Applying Eq. (1) on the muon's own clock gives the fraction that survives the journey,

$$P_{\mathrm{surv}} = \frac{N}{N_0} = \exp\!\left(-\frac{L}{\gamma v \tau_\mu}\right). \tag{2}$$

For the illustrated $L = 15$ km and $v = 0.998c$, $\gamma \approx 15.8$. The flight lasts 50.1 µs in the ground frame but only 3.17 µs for the muon, which sees the distance contracted to about 950 m (Fig. 2). Approximately 24% of the muons survive, compared with about 10^{-10} if time dilation is ignored ($\gamma = 1$). A prediction of the actual flux would also need the distributions of production height and energy, so the example only shows the size of the effect.

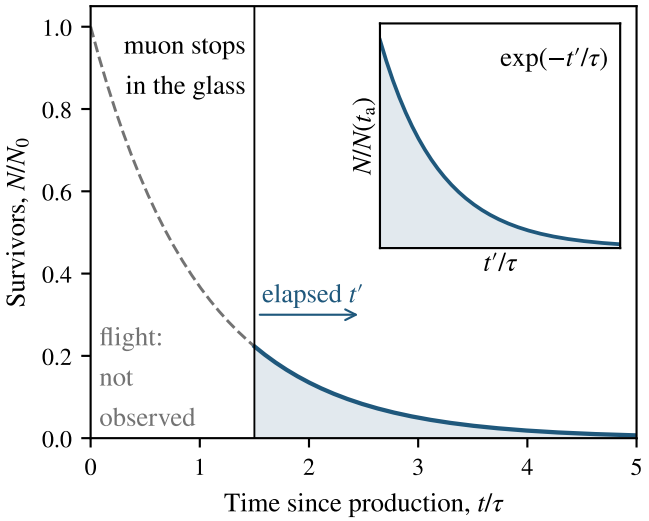


FIG. 3. Why the waiting time after stopping measures the full lifetime. The curve is the surviving fraction of a muon population against time since production. The flight through the atmosphere (dashed) is not observed; the muon comes to rest in the glass at age $t_a = 1.5\tau$, which defines the beginning of its rest-frame waiting time t' (shaded). The electronic START follows after a short delay, described with the apparatus. Inset: the recorded part, shifted to start at zero and divided by its value at stopping, is $\exp(-t'/\tau)$, the same curve as the population since production, whatever the age at stopping (Eq. (3)). Illustrative calculation; no data are shown.

2.3 What the detector measures

Only the low-energy part of the incident flux stops inside the glass, because each muon loses kinetic energy steadily by ionization along its path. A veto counter below the block rejects many of the muons that pass straight through. Once a muon has stopped, its remaining waiting time is measured in its own rest frame, so no time-dilation factor applies to the recorded interval.

How long each stopped muon spent in flight is unknown, and the apparatus does not measure it. It records only the waiting time t' between the muon stopping at age t_a and its decay. Of the muons that survive to t_a, the fraction still present a time t' later follows from Eq. (1):

$$\frac{N(t_a + t')}{N(t_a)} = \frac{N_0 \exp(-(t_a + t')/\tau)}{N_0 \exp(-t_a/\tau)} = \exp\!\left(-\frac{t'}{\tau}\right). \tag{3}$$

The unknown age cancels. The waiting time falls exponentially with the same τ as the population since production, so the measured part of the curve has the same shape as the whole (Fig. 3). Because this holds for every muon, waiting times from muons of many different ages can be collected in one histogram, and their ages only set how many survive long enough to be stopped. Each recorded interval is one sample from the density $p(t') = \exp(-t'/\tau)/\tau$, so later bins of the histogram hold fewer counts.

The decay electron is seen through the light it produces in the glass. Lead glass emits Cherenkov radiation when a charged particle moves faster than light travels in it, $\beta > 1/n$, where n is the refractive index [5]. This is analogous to a sonic boom: the particle outruns the propagation of light in the medium,

as a supersonic object outruns sound. It does not exceed the vacuum speed of light. Setting $\beta = 1/n$ in the Lorentz factor gives the threshold $\gamma_{\mathrm{th}} = (1 - n^{-2})^{-1/2}$, and since the kinetic energy is $K = (\gamma - 1)mc^2$ for a particle of mass m,

$$K_{\mathrm{th}} = mc^2\Big(\big(1 - n^{-2}\big)^{-1/2} - 1\Big). \tag{4}$$

For $n = 1.62$, the value quoted for a similar block [6], the threshold is about 0.14 MeV for an electron and 29 MeV for a muon. As a result, an incoming muon radiates while it is fast and falls silent as it slows and stops, and the energetic decay electron later produces a new flash [6]. Some decay electrons escape the block or give pulses below the discriminator threshold. These losses reduce the number of recorded decays but leave the slope unchanged, provided the detection efficiency does not depend on the waiting time.

2.4 Competing nuclear capture

Nuclear capture affects only negative muons. A stopped negative muon falls into an atomic orbit close to a nucleus, where it can be absorbed, whereas a positive muon is repelled. Capture and decay are independent ways to disappear, so their probabilities per unit time add. Taking the decay rate of a bound muon equal to the free rate $1/\tau_\mu$, a negative muon stopped in element j disappears with time constant τ_j given by

$$\frac{1}{\tau_j} = \frac{1}{\tau_\mu} + \Lambda_j, \tag{5}$$

where Λ_j is the capture rate [7]. Capture gives negative muons an additional way to disappear, so their lifetime is shorter, or equivalently their disappearance rate is higher, than for free positive muons. The detected sample contains both charges and negative muons bound to different elements. Fitting this mixture with one exponential therefore gives an effective lifetime whose shift depends on the mixture and the retained part of the spectrum.

3. EXPERIMENTAL METHOD

3.1 How the instruments form a time interval

The apparatus turns two flashes of light into one recorded time interval (Fig. 4). Two photomultiplier tubes (PMTs), left and right, view the lead glass. A scintillation counter above the block marks a muon's arrival, and a counter below it vetoes muons that pass straight through.

In each PMT, photons release electrons from the photocathode. The high voltage accelerates them through dynodes that multiply their number, producing a measurable pulse at the anode (Fig. 5). Electrons in the cascade can also ionize residual gas; returning ions may release another burst from the photocathode [8]. The delay depends on the ion's mass, from about 0.3 µs for H_2^+ to a few microseconds for heavier ions [9]. Such a delayed pulse could stop the timer before the true decay. The narrow excess holds roughly 6–16% of recorded intervals and is excluded from the fit. Tests of the remaining shape and window stability determine which runs support a

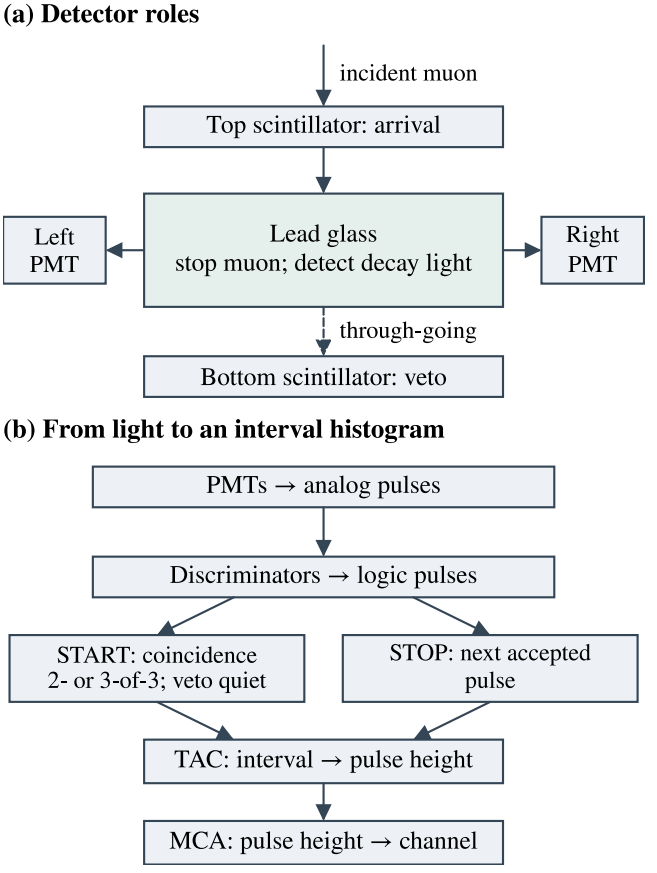


FIG. 4. Detector roles and signal flow. The top counter tags arrival, the glass PMTs detect light and the bottom counter vetoes through-going particles. "Veto quiet" means no bottom-counter pulse, not that the veto is disabled. Discriminated pulses drive the coincidence logic; the time-to-amplitude converter (TAC) turns an accepted interval into a pulse height and the multichannel analyzer (MCA) stores it as a channel. Geometry is schematic.

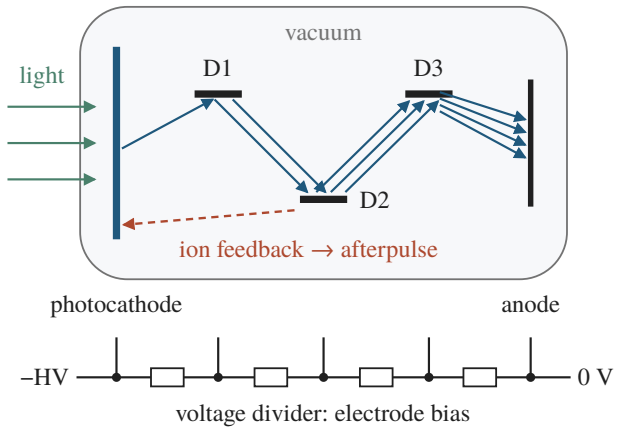


FIG. 5. Photoemission and electron multiplication in a PMT. Light releases photoelectrons at the photocathode, and each dynode multiplies them before the anode collects the pulse; the divider below sets the electrode voltages. An ion created in the cascade can drift back to the photocathode (red, dashed) and release a delayed afterpulse. Dynode count and geometry are illustrative [8].

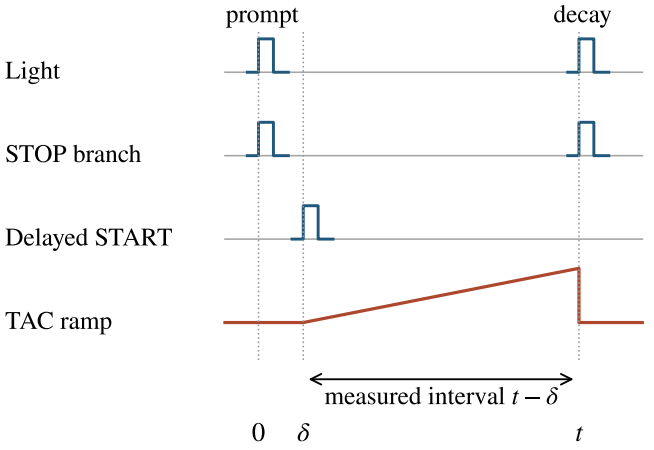


FIG. 6. Idealized delayed-START sequence; horizontal separations are not measured delays. The prompt STOP precedes the delayed START and is ignored. The decay pulse ends the ramp after $t - \delta$, where t is the waiting time and δ the net START delay.

lifetime estimate (Sec. 4.2); exclusion alone cannot recover a later decay lost to a false STOP.

The PMT pulses are turned into a time measurement in several stages. A discriminator emits a standard logic pulse whenever its input crosses a threshold [10]. Raising the threshold rejects more electronic noise but also loses weak decay pulses. A coincidence unit then requires two or three of the left, right and top signals to overlap while the bottom veto is quiet [11]. Accepted START and STOP signals drive a time-to-amplitude converter (TAC), which charges a capacitor at constant current between them, so its output pulse height is proportional to the interval [12]. The multichannel analyzer (MCA) sorts these heights into 8192 channels. To convert channel number into time, a calibrator supplies START–STOP pairs separated by multiples of a crystal-controlled period [13].

Because the incoming muon also produces light in the glass, it would stop the timer immediately unless the START is delayed until this prompt pulse has passed (Fig. 6). The next accepted pulse then ends the interval.

A fixed delay δ shortens every recorded interval to $t_{\mathrm{meas}} = t - \delta$. Substituting into the exponential,

$$\exp\left(-\frac{t}{\tau}\right) = \exp\left(-\frac{\delta}{\tau}\right)\exp\left(-\frac{t_{\mathrm{meas}}}{\tau}\right), \tag{6}$$

which changes only the amplitude, through the same cancellation as the unknown muon age in Eq. (3). No correction for the delay is applied to the fitted lifetime.

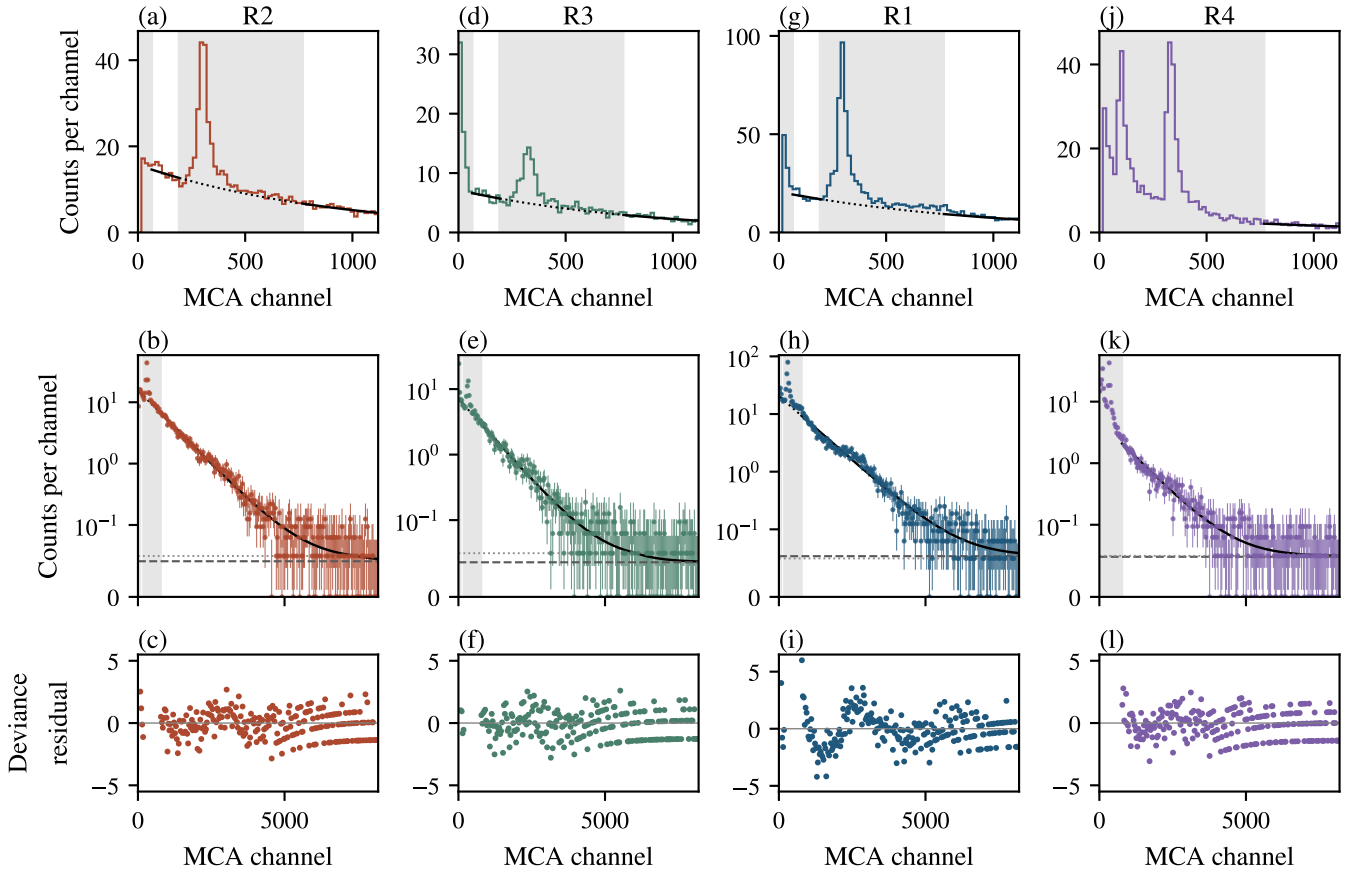


FIG. 7. Spectra and exponential-plus-constant fits for R2 (a–c), R3 (d–f), R1 (g–i) and R4 (j–l). Shading marks excluded channels; dotted curves extrapolate across the gap. Fits retain 64–192 and 768–8192, or only 768–8192 for R4. Top: early spectrum in 16-channel bins, showing the secondary peaks. Middle: full range in 32-channel bins with 68% Poisson bars and fitted background (dashed). Both rows show counts per channel; the middle axis becomes linear below one count per bin (dotted line). Bottom: signed deviance residuals; discrete late-time bands arise from zero- and one-count bins. Only R2/R3 pass both shape and window-stability tests.

High voltages, thresholds, coincidence level and the veto cable length were adjusted between the four acquisitions, and not all of their values were recorded. Each spectrum was therefore fitted on its own.

3.2 Data and fitting around the afterpulse

Four acquisitions R1–R4 contain 26,358, 16,435, 7,797 and 11,053 intervals from 117, 120, 42 and 115 h of live time, or 224, 137, 185 and 96 intervals per hour. Changed thresholds and coincidence requirements alter pulse acceptance, so a higher rate need not mean a better lifetime measurement. Each spectrum was fitted separately without smoothing or background subtraction.

In Fig. 7, a channel is a bin in TAC output pulse height and hence in the recorded waiting time: larger channel numbers mean longer intervals. Below channel 64, a sharp rising edge is inconsistent with a smoothly falling exponential. Prompt-pulse leakage or timing acceptance could cause it; its origin is unresolved. This region is excluded and the boundary varied to test sensitivity. A narrow secondary peak between about 192 and 500 is tentatively attributed to afterpulsing. The fit retains an early interval, 64–192, and a tail, 768–8192, spanning the gap with one exponential. The later tail start avoids the strongest initial drift (Sec. 4.2). R4 has an extra peak near 100, so its early interval is excluded too.

The model combines the decay law of Sec. 2.2 with a background. The time scale is linear, $t = \alpha c + t_0$, where c is the channel and α the time per channel. By the same cancellation as in Eq. (6), the decay $\exp(-t/\tau)$ becomes $\exp(-c/\ell)$ in channel number, with decay length $\ell = \tau/\alpha$; the offset t_0 only rescales the amplitude. Accidental coincidences, STOP pulses unrelated to the muon that started the timer, arrive at random times; at low rates they spread almost evenly over the range and add a constant background. Neighbouring channels were summed into bins of 32, and the expected count in the bin from c_i to c_{i+1} is the integral of this density,

$$m_i = \int_{c_i}^{c_{i+1}} \left(a \exp\left(-\frac{c - c_0}{\ell} \right) + b \right) \mathrm{d}c, \tag{7}$$

where a is the decay density at the first fitted channel c_0 and b the background density. Bins outside the windows simply do not enter the fit, so one continuous exponential spans the gap. The parameters were found by maximising the Poisson likelihood with $a, b \geq 0$, and $\tau = \alpha\ell$. Counts in a bin follow Poisson statistics [14]. Least squares corresponds to maximum likelihood for Gaussian errors; R2, for example, has 31 empty fitted bins, so fitting log counts or assigning errors $\sqrt{y_i}$ would mishandle the tail.

Three quantities describe each fit. The 68% statistical interval on ℓ comes from profiling the likelihood over the other parameters and raising its negative logarithm by 1/2 [15]. Fit quality is measured by the Poisson deviance, twice the logarithm of the likelihood ratio between a perfect fit ($m_i = y_i$) and the fitted model [16],

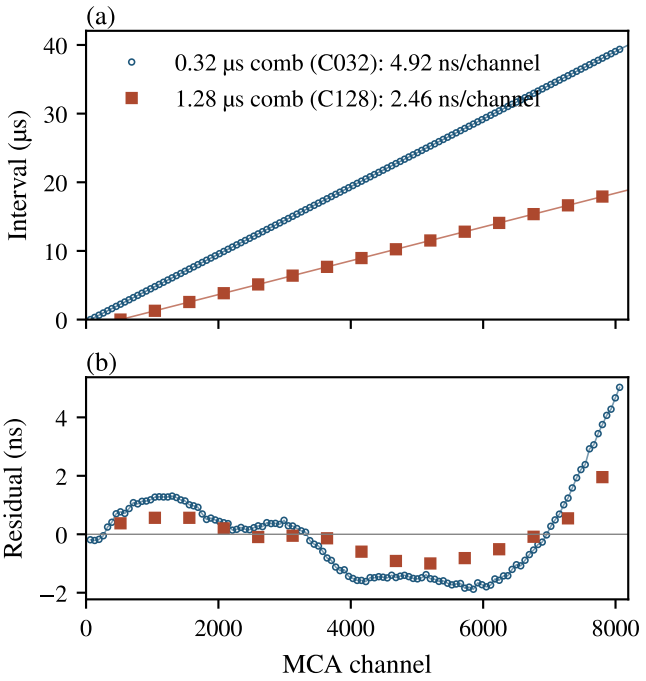


FIG. 8. Calibration of channel number to time. (a) Known pulse separations against centroid channel, with linear fits. Each set of evenly spaced peaks forms a calibration "comb." C128 uses the 1.28 µs period used in the experimental calibration; C032 uses its file-labelled period. (b) Residuals from the straight lines.

$$D = 2 \sum_i \left(y_i \ln\left(\frac{y_i}{m_i} \right) - (y_i - m_i) \right), \tag{8}$$

where y_i is the observed count and the logarithm is taken as zero when $y_i = 0$. For large expected counts, D approaches the familiar $\sum_i \frac{(y_i - m_i)^2}{m_i}$, with Poisson variance m_i. It is therefore the counting-data counterpart of the uncertainty-weighted χ^2. The sparse tail makes that approximation unreliable; its probability was instead found from 399 simulated spectra generated from each fitted model and fitted in the same way. Tail stability was tested at seven starts from 768 to 2048, reusing each simulated histogram across cuts to retain their correlations. Only runs with both deviance and stability probabilities at least 0.05 enter the combination.

With k simulated exceedances, the reported probability is $\frac{k+1}{400}$; 0.0025 denotes simulation resolution, not a confidence bound. These tests condition on the fitted model and fixed cuts, without covering the data-informed choice of 768.

4. RESULTS AND ANALYSIS

4.1 Timing scale and calibration transfer

The TAC and analyser must be calibrated together to turn a channel number into a waiting time. A calibrator supplies START–STOP pairs separated by known multiples of its period. These repeated intervals produce evenly spaced peaks, called a calibration comb (Fig. 8). Fitting assigned time against peak channel gives the conversion α in time per channel. Which

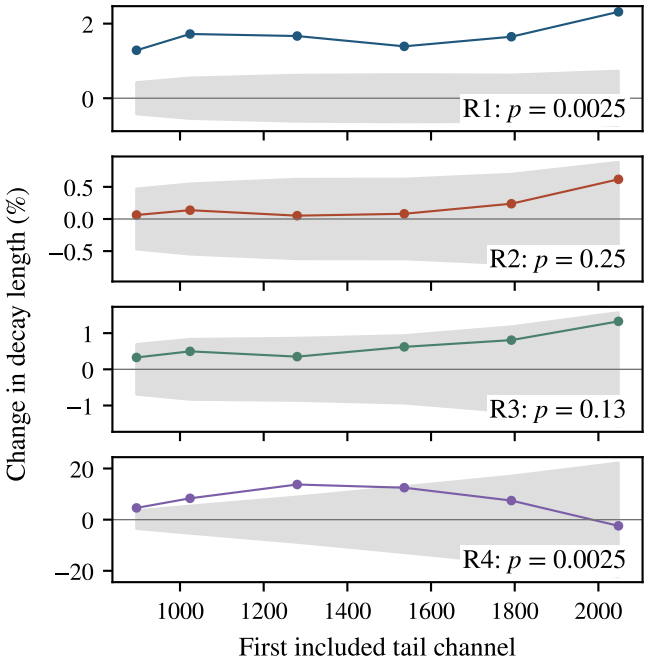


FIG. 9. Change in decay length as the tail starts later, relative to channel 768, in percent; each panel has its own vertical scale so that every envelope stays visible. The early window remains in R1–R3; R4 uses the tail alone. Gray bands are 95% simultaneous envelopes from the fitted single-exponential model. Each simulated spectrum is reused at every cut. The probabilities describe the largest standardized departure over the tested cuts; 0.0025 denotes zero exceedances in 399 trials.

interval produced the first visible peak changes only the offset, not the slope or a single-exponential lifetime.

The apparatus was manually calibrated with the 1.28 µs setting, as recalled by the author and used in the partner's analysis. That calibration, C128, is used here. Its 15 peaks give $\alpha = 2.4609$ ns per channel, with 0.73 ns rms residuals. The comb labelled 0.32 µs gives 4.9214 ns per channel, twice the C128 slope to within 1 part in 10^4, and would give lifetimes near 4.44 µs, which no muon process allows. The TAC offers only discrete full-scale ranges (50, 100 or 200 ns times a power of ten) [12]: C128 matches the 20 µs range to 1%, whereas the 0.32 µs label would need a 40 µs range that does not exist. Both points support C128 independently of the recalled setting.

The remaining question is whether the scale stayed unchanged between acquisitions. Both combs and R3 were recorded on input 2, while R1, R2 and R4 used input 1; no before-and-after calibration record was supplied for each configuration. The recalled procedure motivates adopting C128, but does not verify its transfer to each input and run. The tested smooth nonlinearity is small: a quadratic calibration changes the fitted lifetimes by less than 0.04%.

4.2 Tail stability and run selection

Sidebands put R2's main peak at 9.7–10.5% of recorded intervals; R4's extra early peak holds 8.6%.

A single exponential keeps its slope when the first fitted channel moves later; residual peak contamination could change it. The cut test asks whether the observed change exceeds

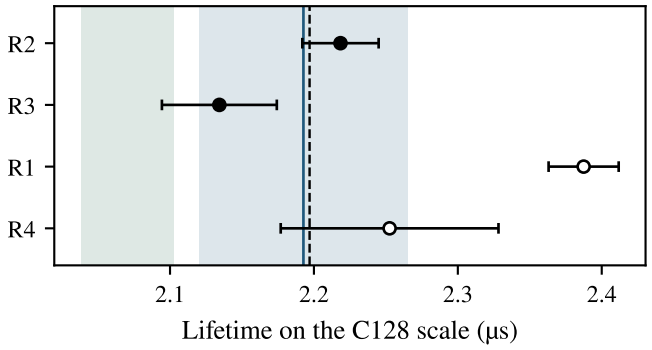


FIG. 10. Four-run lifetime comparison on the C128 scale. Bars are statistical intervals; filled points pass both diagnostics, open points fail at least one. The solid line and blue band show the combined value and its statistical-plus-scatter and systematic terms added in quadrature. The green band spans two illustrative capture-model predictions, not a confidence interval; the dashed line is the free-muon lifetime.

the fluctuation expected from discarding counts. Reusing one simulated histogram across cuts accounts for the shared events.

Moving its start from 512 to 768 increases the fitted decay length by 3.2%, 1.0%, 1.2% and 12.1% in R1–R4. The common start at 768 was chosen after examining this behaviour.

Beyond channel 768, R2 and R3 pass the paired-cut test (Fig. 9), with probabilities 0.25 and 0.13. Their decay lengths still change by 0.6% and 1.3% between 768 and 2048; this residual sensitivity is retained in the uncertainty budget. R1 and R4 fail the same test, each with zero exceedances among 399 simulations. Their late-cut drift is therefore not explained by counting fluctuations under the fitted model, even when an individual window gives an acceptable deviance.

The comparison uses one set of rules for all four acquisitions: the nominal deviance and paired-cut probabilities must both be at least 0.05. Only R2, R3 pass.

4.3 Decay fits on both sides of the afterpulse

R2 and R3 give the two usable estimates in Fig. 10. R2 has decay length $\ell = 901$ channels, with a 68% profile interval of 891–912 channels. Its deviance is 262.8 for 233 nominal degrees of freedom, with simulated probability 0.29. The fit and residuals are shown in Fig. 7 a–c. On C128 this is (2.22 ± 0.03) µs, where the error includes counting statistics only. R3 gives $\ell = 867$ channels (851–884), or (2.13 ± 0.04) µs, with deviance probability 0.11 (Fig. 7 d–f).

R1 has more counts but fails the shape test: its deviance exceeds that of every simulated spectrum. Its fitted value, (2.39 ± 0.02) µs, is shown to expose the disagreement, not used as a valid lifetime estimate. The residuals in Fig. 7 i dip between channels 1300 and 2300 and rise between 2300 and 3300, a shape a single exponential cannot follow. R4 gives (2.25 ± 0.08) µs from its tail alone, with nominal deviance probability 0.06. Some later windows are individually acceptable and give 2.20–2.44 µs, but the paired-cut test rejects the stability of its fitted length. R1 and R4 are therefore not combined.

R2 and R3 differ by 1.8 combined statistical standard deviations despite individually acceptable fits. The combination

TABLE I. Uncertainty and sensitivity contributions in µs. Each source is defined in the text below; sensitivities are not confidence limits. Systematic totals combine the listed symmetric sources in quadrature. The final two rows are separate limitations and are not included in that total.

Source	R2	R3	Combined
Counting	0.027	0.040	0.022
Background	0.044	0.039	0.042
Tail-cut drift	0.007	0.014	0.009
Early acceptance	0.052	0.023	0.043
Binning	0.001	0.001	0.001
Slope regression	< 0.001	< 0.001	< 0.001
Systematic total	0.069	0.047	0.061
Configuration spread			0.032
Accidental STOPs	≤ 0.003	≤ 0.006	
Scale transfer (inputs)	unknown	unknown	unknown

allows for this configuration-dependent spread without identifying its cause.

4.4 Uncertainty and combined result

The budget in Table I distinguishes statistical precision from sensitivity to the model and windows. Allowing independent exponential amplitudes before and after the gap gives early-to-tail ratios of 1.08 ± 0.04 for R2 and 1.03 ± 0.06 for R3. R2′s length becomes 923 channels, outside the nominal 68% interval although the two profile intervals overlap. R2 has a modest early excess, about two standard errors above unity; R3 is consistent with unity. The separate-amplitude shifts are 0.052 and 0.023 µs. The early-acceptance term is the larger of this shift and the half-range from edge scans (starts 48–128, ends 160–208), avoiding double-counting these related probes.

The residual tail-cut range from 768 to 2048 contributes another half-range. Replacing the constant background by a nonnegative linear one changes R2′s length to 888 channels; the improvement has simulated probability 0.165. Ending the fit at 4096 instead of 8192 leaves fewer late bins to fix the background level and shifts the lengths by 884–901 channels for R2. Both probes test the same background freedom, so the larger of the two shifts is taken as a symmetric sensitivity. Binning and calibration-regression terms complete the budget. These choices use the same events and are not strictly independent, so quadrature is an approximate summary rather than a calibrated confidence interval.

The combination is an inverse-variance weighted mean [17], with the independent counting errors supplying weights (R2 0.69, R3 0.31). Shared systematic effects are kept separate, as their shifts do not average down with independent counts. Its mean is 2.19 µs, with unscaled standard error 0.022 µs and $\chi^2 = 3.1$ for 1 degree of freedom. In the manner of the Particle Data Group's scale factor [2], but computed from the counting errors alone, the standard error is multiplied by $S = \max\left(1, \sqrt{\frac{\chi^2}{N-1}}\right) = 1.8$, giving 0.039 µs. Equivalently, an extra 0.032 µs is added in quadrature to the unscaled error to represent the observed spread.

Each systematic source is conservatively assumed fully positively correlated between runs: its weighted contributions are added before combining different sources in quadrature. This gives 0.061 µs. The result is

$$\tau = (2.19 \pm 0.04\ (\text{stat} + \text{scatter}) \pm 0.06\ (\text{sys}))\ \mu\text{s}.$$

With only two selected acquisitions, the scatter allowance is poorly determined. Weighting each run by its total error instead gives 2.17 µs, a measure of how much the central value depends on the weighting choice. The result uses C128 and excludes unquantified changes in that scale between configurations. The accidental-STOP estimate is separate because it is one-sided and model-dependent (Sec. 4.5).

4.5 Accidental stops and a capture estimate

Accidental STOPs arriving independently at rate r compete with decay, giving $\tau_{\text{obs}} = \frac{\tau}{1+r\tau}$. The fitted flat background bounds their rate if acceptance is constant and the number of STARTs is at least the recorded count. This gives possible shortenings of 0.003 µs for R2 and 0.006 µs for R3, small beside Table I.

The capture prediction was calculated here to estimate the shift from the free lifetime. The assumed glass is 45% PbO and 46% SiO_2 by mass, with the unknown 9% omitted and retained shares renormalized. The assumed stopped charge ratio is 1.27 [18], and atomic capture shares follow the approximate Fermi–Teller rule, atomic number times atom count [19]. Literature negative-muon lifetimes for Pb, Si and O set survival; the electron-producing decay rate sets amplitude [7].

The resulting expected electron counts, with R2′s background fraction, were fitted using the same masked model as the data. Assigning the first calibrator peak to one or two periods gives two illustrative origins and fitted lifetimes of 2.04 and 2.10 µs. The START delay was not measured, so this is a range of scenarios, not a measurement. It lies 0.09–0.15 µs below the measured mean (Fig. 10). Predicted early-to-tail ratios of 1.04–1.12 overlap the small observed excesses and do not reject the illustration.

5. DISCUSSION

5.1 What the spectra establish

The retained spectra support an exponential decay scale, but the secondary peaks show that a recorded STOP is not always a decay. PMT afterpulsing is a plausible explanation: a large prompt pulse can generate a delayed pulse at a preferred time, unlike the smoothly falling decay distribution. R4′s second peak could reflect another ion species in the tube, since ions of different mass return after different delays [9]. Cable reflections or logic effects remain alternatives, so the attribution is tentative until pulse traces establish the mechanism.

Removing the peak is necessary but not sufficient. A false STOP may replace a later decay rather than merely add an extra count. If every afterpulse STOP did so, the tail would be suppressed relative to the early window, giving early-to-tail ratios of 1.20, 1.16 and 1.10 for R1–R3 from the peak areas. The measured ratios, 1.04 ± 0.03, 1.08 ± 0.04 and 1.03 ± 0.06, are smaller, so most afterpulse STOPs must follow STARTs

that had no detected decay, such as muons captured in lead or decay electrons that escaped. The tail-cut and amplitude tests ask whether the retained spectrum is still distorted; their outcome supports using R2 and R3 while excluding R1 and R4.

The capture comparison also limits the interpretation of the lifetime. The model in Sec. 4.5 is a calculation from assumed material properties, not an independently measured correction for this block. Its range lies below the measured mean, but the small early excess is compatible with its predictions. Therefore proximity to the free-muon lifetime does not by itself show that capture is negligible or that the apparatus measures only positive-muon decay.

The recalled calibration setting, supported by the TAC-range argument of Sec. 4.1, justifies using C128. What remains unmeasured is its repeatability under the changing run settings. More counts would improve the precision of each conditional fit, but would not determine the origin of the secondary peaks, establish a stable detection efficiency or check the scale between configurations. Those are the measurements needed to turn the effective decay scale into a more precise physical lifetime.

5.2 Measurements that would reduce the limitations

Recording a calibration comb on the same analyzer input, with the TAC range and gain unchanged, before and after each acquisition would fix the time scale directly and reveal any drift.

The afterpulse origin can be tested by viewing the prompt and delayed PMT pulses on an oscilloscope, and by changing one setting at a time: a peak that stays fixed when cable lengths change but moves with PMT voltage would support an origin in the tube. Raising a PMT threshold or inhibiting STOPs over the secondary-peak delay could suppress false stops, but would also change decay acceptance. That acceptance must be checked before extending the fitted interval.

Recording the number of STARTs would turn the accidental-rate bound into a measurement. The composition of the glass, or a comparison with a block of known composition, would turn the capture estimate into a correction. Repeating acquisitions with fixed settings would separate statistical scatter from configuration-dependent response.

6. CONCLUSION

Two of four stopped-muon acquisitions support a stable single-exponential description after the secondary peaks and start-edge structure are excluded. Their combination gives $\tau = (2.19 \pm 0.04\ (\text{stat + scatter}) \pm 0.06\ (\text{sys}))\ \mu\text{s}$ using the C128 calibration. The first uncertainty includes the disagreement between configurations; the second summarizes sensitivity to the model and fitting windows. Neither establishes a bound on unrecorded calibration drift.

The measured scale is compatible with the free-muon lifetime, while the illustrative capture calculation predicts a value 0.09–0.15 µs shorter. The data cannot yet tell whether the capture assumptions or an untested difference in time scale between analyser inputs is responsible. Its material and detector assumptions are not sufficiently constrained to extract a capture correction. The experiment therefore measures an effective decay scale whose interpretation is limited by detector response. Fixed-setting repeats, matched calibrations and direct pulse measurements would address those limitations more effectively than additional counting time alone.

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REFERENCES

[1] V. Tishchenko et al., Detailed report of the MuLan measurement of the positive muon lifetime and determination of the Fermi constant, Physical Review D **87**, 052003 (2013).

[2] S. Navas et al. (Particle Data Group), Review of Particle Physics, Physical Review D **110**, 030001 (2024).

[3] A. C. Melissinos and J. Napolitano, *Experiments in Modern Physics*, 2nd ed. (Academic Press, 2003).

[4] D. H. Frisch and J. H. Smith, Measurement of the Relativistic Time Dilation Using µ-Mesons, American Journal of Physics **31**, 342 (1963).

[5] D. E. Groom and S. R. Klein, *Passage of Particles Through Matter*, Review of Particle Physics, Particle Data Group, revised August 2023, https://pdg.lbl.gov/2024/reviews/rpp2024-rev-passage-particles-matter.pdf.

[6] B. Dickson, *Determination of Mean Muon Lifetime* (PHY 474 student laboratory report, The University of Texas at Austin, 2014).

[7] T. Suzuki, D. F. Measday, and J. P. Roalsvig, Total nuclear capture rates for negative muons, Physical Review C **35**, 2212 (1987).

[8] Hamamatsu Photonics, *Photomultiplier Tubes and Related Products*, Catalogue TPMZ1036E, section on ion feedback, https://www.hamamatsu.com/content/dam/hamamatsu-photonics/sites/documents/99_SALES_LIBRARY/etd/PMT_TPMZ1036E.pdf.

[9] PHOTONIS, *Photomultiplier Tubes Catalogue, Part 2: Photomultiplier Tubes Basics* (PHOTONIS Group, 2007).

[10] Phillips Scientific, *Model 705 Octal Discriminator: Data Sheet* (Phillips Scientific, Mahwah, NJ, n.d.).

[11] Phillips Scientific, *Model 755 Quad Majority Logic Unit: Data Sheet* (Phillips Scientific, Mahwah, NJ, n.d.).

[12] ORTEC, *Model 566 Time-to-Amplitude Converter: Operating and Service Manual, Part No. 678950, Rev. D* (ORTEC, Oak Ridge, TN, n.d.).

[13] ORTEC, *Model 462 Time Calibrator: Operating and Service Manual, Part No. 733340, Rev. G* (ORTEC, Oak Ridge, TN, n.d.).

[14] PHY 474 course materials, *Statistical Analysis, Parts 1–3* (Lecture slides, Advanced Laboratory I, The University of Texas at Austin, 2026).

[15] P. R. Bevington and D. K. Robinson, *Data Reduction and Error Analysis for the Physical Sciences*, 3rd ed. (McGraw-Hill, 2003).

[16] S. Baker and R. D. Cousins, Clarification of the use of chi-square and likelihood functions in fits to histograms, Nuclear Instruments and Methods in Physics Research **221**, 437 (1984).

[17] PHY 474 course materials, *Measurement Uncertainties* (Lecture slides, Advanced Laboratory I, The University of Texas at Austin, 2026).

[18] J. J. Beatty, J. Matthews, and S. P. Wakely, *Cosmic Rays*, Review of Particle Physics, Particle Data Group, revised October 2021; muon charge ratio in Fig. 30.7, https://pdg.lbl.gov/2022/reviews/rpp2022-rev-cosmic-rays.pdf.

[19] E. Fermi and E. Teller, The Capture of Negative Mesotrons in Matter, Physical Review **72**, 399 (1947).